

Challenge (Habanero)

- Q1.** A gamer scored $\frac{3}{5}$ of the total points in a game. If the total points are 5000, how many points did the gamer score?
(A) 3000
(B) 2500
(C) 2000
(D) 1500
(E) 1000
- Q2.** A new smartphone's battery life is $2\frac{1}{4}$ hours when playing games. How many hours is this in decimal form?
(A) 2.25
(B) 2.40
(C) 2.50
(D) 2.10
(E) 2.75
- Q3.** A game developer wants to increase the speed of a character by 15%. If the current speed is 25 units per second, what is the new speed?
(A) 28.75
(B) 29.25
(C) 28.00
(D) 27.50
(E) 27.00
- Q4.** A gamer has $\frac{7}{10}$ of a dollar to spend on in-game items. How much is this in cents?
(A) 70
(B) 65
(C) 75
(D) 60
(E) 50
- Q5.** A coding error causes a game to use $1\frac{3}{5}$ times more data than usual. If the usual data usage is 20 MB, how much data is used now?
(A) 36
(B) 40
(C) 38
(D) 32
(E) 46
- Q6.** A scoreboard shows a team's score as 0.45 goals per minute. How many goals will they score in $2\frac{1}{2}$ minutes?
(A) 1.125
(B) 1.100
(C) 1.150
(D) 1.200
(E) 1.250
- Q7.** A gamer's high score is $\frac{9}{20}$ of the world record. If the world record is 10000 points, how many points is the gamer's high score?
(A) 4500
(B) 5000
(C) 4000
(D) 3500
(E) 3000
- Q8.** A game uses $3\frac{1}{8}$ GB of storage space. How many MB is this?
(A) 3008
(B) 3072
(C) 3080
(D) 3096
(E) 3104

Open Ended Questions (FRQ)

- Q9.** A new game requires $\frac{3}{8}$ GB of free disk space to install. If a gamer has $2\frac{1}{4}$ GB of free space, will they have enough space to install the game? Show your work and explain your answer.
- Q10.** A gamer's favorite game has a new update that increases the game's file size by 20%. If the original file size is 5 GB, how many GB will the updated game be? Show your work and explain your answer.

Chef's Tips

- Tip 1: Emphasize the importance of converting between fractions, decimals, and percents
- Tip 2: Encourage students to use number lines to compare and order rational numbers
- Tip 3: Remind students to check their units when performing calculations with different units